

DAT565/DIT407 Assignment 1

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Report of assignment1 for the course *Introduction to Data Science & AI*.

Problem: Dependency ratio

The purpose of this task is to use Python tools like Pandas, NumPy, and Matplotlib to analyze the population structure of Sweden. These packages are used to construct a figure that displays Sweden's dependency ratio from 1860 to 2022 and read the dataset. This statistic helps to understand the economic and social pressures on the national economy of a country.

A demographic measure called the dependency ratio provides information on how the population is divided between productive and dependent populations. The dependent group consists of the elderly and children, whereas the productive group is made up of labor forces [3].

Data Processing

We initiated by importing the dataset of file `swedish_population_by_year_and_sex_1860-2022.csv` from SCB [2]. In fact, this dataset contains data on the population by age and sex over a specific period. To guarantee numerical consistency, we processed the age column. Specifically, we converted ages indicated with a '+' to their corresponding numerical values.

Grouping the Population

We divided the population into three groups as follows based on the classifications of the World Health Organization (WHO) [3]:

- Children (0–14 years)
- Labor force (15–64 years)
- Elderly (65 years and above)

These categories allow us to compute the dependency ratio precisely since it is based on how big these groups are in compare to others. As the dependency ratio is based on age, so gender is not considered in our computation.

Calculating the Dependency Ratio

After classifying the data, the ratio percentage is computed by dividing the total sum of children and elderly by the working population and then multiplying the result by 100. An overview of the demographic pressure on the productive population group can be seen by this ratio [3].

$$\text{Dependency ratio} = 100 * \frac{\#children + \#elderly}{\#labor\ force} \quad (1)$$

Data Visualization

We generated 2 graphs by Matplotlib, which highlights how population loads impact Sweden's socioeconomic structure and how they have shifted over time. One illustrates the dependency ratio from 1860 to 2022 and the other one shows the fraction of the children, the elderly, and the dependency ratio in the same period.

As we can see in Figure 1 and Figure 2, examining population statistics shows that Sweden's demographic view has changed significantly over time. These changes could be due to the Industrial Revolution, an increase in life expectancy, declining fertility, expansion of health care, and immigration in different times [1].

At first sight, there has been a decline in the reliance ratio overall during this time. A closer look at the details, shows the maximum turning point, or highest dependency ratio, occurred between 1890 and 1900. The ratio then falls until it reaches the minimum turning point or its lowest in 1940. In recent years, it has begun to grow again, indicating that Sweden's population is getting older.

When the dependency ratio is compared to the fractions in the second plot, it shows that fewer people are in their youth, which may be a sign that the fertility rate has dropped and that more people are of working age. The dependency ratio rises as the proportion of the population reaches working age and the proportion of the elderly rises in line with declining fertility rates [1].

Due drop in fertility, there were fewer children which raised the average age of the population. Women could work outside the home, because of fewer family sizes. Fertility levels were still above replacement, leading to substantial population growth, with each generation still larger than the preceding one [1].

Following World War II, population expansion and robust economic growth made it possible to implement welfare reforms, including changes to the pension system and other areas. When the number of people entering the workforce declines and the senior population's declining death rate takes over, a favorable demographic situation like this one comes to an end [1].

Because people are living longer on average these days, most developed countries have raised their retirement age. In their late 50s and beyond, people are

often healthier and more productive. In the industrialized world, the legal retirement age is rising in many countries. [4].

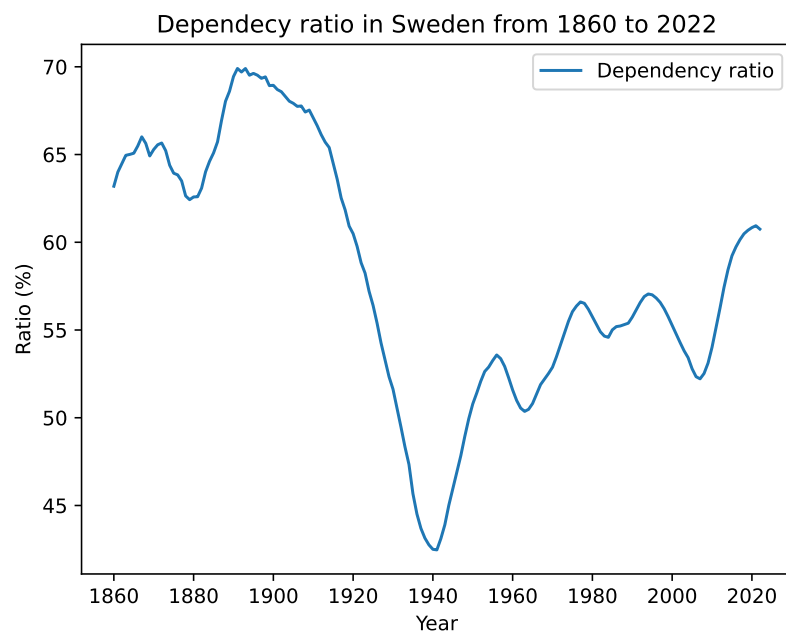


Figure 1: Dependency Ratio in Sweden from 1860 to 2022.

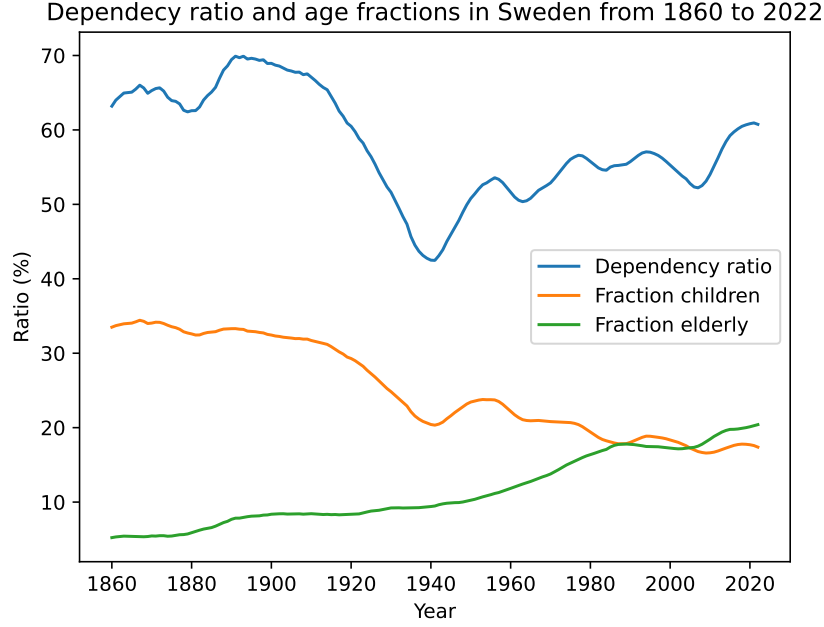


Figure 2: Dependency Ratio and Fraction of children and elderly in Sweden from 1860 to 2022.

To provide a more thorough analysis of the data for recent years, we have also included Table 1, which indicates that Sweden’s ratio was, on average, 55.12 percent throughout from 1960 to 2022, with minimum values of 50.27 percent in 1964 and maximum values of 60.9 percent in 2022. The most current estimate for 2022 is 60.9 percent. However, the global average in 2022, based on 196 countries, is 58.67 percent. [5].

Table 1: Percentages of the dependency ratio in Sweden	
Sweden	Dependent people as percent of the working-age population
Latest value	60,90
Year	2022
Measure	percent
Data availability	1960-2022
Average	55,12
Min-Max	50,27 - 60,9

The demographic changes in Sweden also reflect more general trends in developed countries. Changes in the age distribution of the population, such as a decline in birth rates and an increase in life expectancy, are examples of a population that is aging. Sweden has probably gone through demographic shifts

like many other industrialized countries by improving the healthcare system, changes in family structures, and expansion of the economy [1].

Conclusion

This study plots Sweden's historical dependence ratio using data analysis and visualization, providing insight into the opportunities and difficulties that will arise from future demographic shifts. Because Sweden, like many other countries, is struggling with an aging population, this is important for politicians, economists, and the general public as they navigate the challenges of sustainable development and welfare provision.

References

- [1] T. Bengtsson and K. Scott. *Population Aging and the Future of the Welfare State: The Example of Sweden*. Retrieved 2024-03-23. 2011. URL: <https://onlinelibrary.wiley.com/doi/epdf/10.1111/j.1728-4457.2011.00382.x>.
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- [3] World Health Organization. *The Global Health Observatory Indicator Metadata Registry List: Dependency Ratio*. Retrieved 2023-10-25. 2023. URL: <https://www.who.int/data/gho/indicator-metadata-registry/imr-details/1119>.
- [4] Gustavo De Santis. *Population Ageing in Industrialized Countries: Challenges and Issues*. Retrieved 2024-03-23. 2001. URL: <https://iusssp.org/sites/default/files/prp19.pdf>.
- [5] *Sweden: Age dependency ratio*. Retrieved 2024-03-22. URL: https://www.theglobaleconomy.com/Sweden/Age_dependency_ratio/.

A Sample code

```
1 # This python file takes an csv file about population
2 # as input It calculates the dependency ratio and the
3 # fraction of different age groups over the years and
4 # creates some nice plots
5
6 # Import necessary packages
7 import numpy as np
8 import matplotlib.pyplot as plt
9 import pandas as pd
10
11 # Read in the csv file
12 filename = \
```

```

13     'swedish_population_by_year_and_sex_1860-2022.csv'
14 df = pd.read_csv(filename)
15
16 # Remove plus sign for each element (if any), then
17 # convert into integer
18 df['age'] = \
19     df['age'].apply(lambda x: int(x.replace('+', '')))
20
21 # In order to calculate the dependency ratio for each
22 # year, it is necessary to divide the population for
23 # each year into three groups: the 0 to 14 year old,
24 # the 15 to 64 year olds and the 65+ year olds.
25 # Gender doesn't matter in this case.
26 children = df[(df['age'] < 15)].sum()
27 working_pop = \
28     df[(df['age'] > 14) & (df['age'] < 65)].sum()
29 elderly = df[(df['age'] > 64)].sum()
30
31 # Ignore the first two rows (int of age and sex,
32 # these sums don't make any sense
33 children = children[2:]
34 working_pop = working_pop[2:]
35 elderly = elderly[2:]
36
37 # Calculate the dependency ratio
38 dependency_ratio = \
39     100 * ((children+elderly)/working_pop)
40
41 # Extract years and convert them and the dependency
42 # ratio in np.arrays
43 years = np.array(dependency_ratio.keys())
44 years_array = years.astype('int')
45 dep_array = np.array(dependency_ratio)
46
47 # Plot the dependency ratio over the years
48 fig, ax = plt.subplots()
49 ax.plot(years_array, dep_array,
50         label='Dependency_ratio')
51 plt.title('Dependency_ratio_in_Sweden'
52          'from_1860_to_2022')
53 plt.xlabel('Year')
54 plt.ylabel('Ratio(%)')
55 plt.legend()
56
57 # Save the plot
58 plt.savefig('Dependency_ratio_Sweden.pdf')
59
60 # Calculate the fraction of the children and the
61 # elderly for each year.
62 # Therefore divide the number through the whole

```

```

63 # population (not just working force as for the
64 # dependency ratio)
65 children_frac = \
66     100 * (children/(children+working_pop+elderly))
67 elderly_frac = \
68     100 * (elderly/(children+working_pop+elderly))
69
70 # Convert the fractions into np.arrays
71 children_frac_array = np.array(children_frac)
72 elderly_frac_array = np.array(elderly_frac)
73
74 # Plot the dependency ratio with the fractions of the
75 # children and the elderly
76 fig, ax = plt.subplots()
77 lines = [[dep_array, 'Dependency_ratio'],
78          [children_frac_array, 'Fraction_children'],
79          [elderly_frac_array, 'Fraction_elderly']]
80 for line in lines:
81     ax.plot(years_array, line[0], label=line[1])
82
83 plt.title('Dependency_ratio_and_age_fractions_in_'
84          'Sweden_from_1860_to_2022')
85 plt.xlabel('Year')
86 plt.ylabel('Ratio(%)')
87 plt.legend()
88
89 # Save the plot
90 plt.savefig('Dependency_ratio_and_'
91          'fractions_Sweden.pdf')

```